The RALE Model of a Reserve Bank

A Nonlinear Dynamical Framework for Reserves, Assets, Liabilities, and Equity

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Abstract

We introduce the **RALE model**, a self-referential nonlinear dynamical system that simultaneously characterises the four core quantities of a reserve bank's balance sheet: Reserves $R(t)$, Assets $A(t)$, Liabilities $L(t)$, and Equity $E(t)$. The defining equations are

$$A(t) = R(t)\big(1 + L(t)\big), \quad L(t) = A(t) - E(t), \quad E(t) = E(0)\big(1 + A(t)\big)^t,$$

which reduce to a single implicit fixed-point equation in $A(t)$. Because this equation admits no closed-form solution in general, the model captures the inherently numerical character of central-bank balance-sheet management. We analyse existence and uniqueness of equilibria, derive a bisection algorithm for real-time computation, study the linearised dynamics, and connect the model to the money-multiplier framework, capital-adequacy regulation, and data-driven forecasting via gradient-boosted fixed-point iteration. Numerical experiments illustrate solvency spirals, liquidity crunches, and path-dependent equity accumulation.

The paper ends with "The End"

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1 Introduction

Reserve banking rests on a simple accounting identity: a bank's assets equal its liabilities plus its equity. Yet the *dynamics* that govern how each quantity evolves are far from simple. Reserves constrain asset expansion through regulatory requirements; liabilities grow as deposits are attracted; and equity accumulates through retained earnings that are themselves functions of the asset base. Standard textbook treatments linearise these dependencies, yielding the classical money-multiplier formula $A = R/\rho$ where ρ is the reserve ratio. That formula is exact only in steady state and suppresses all feedback between equity and assets.

This paper proposes a minimal nonlinear model—the **RALE model**—that restores those feedbacks without introducing excessive parameter complexity. The model is defined by three simultaneous equations (Section 2) whose combination produces a scalar implicit equation whose unique solution must be obtained numerically. We argue that this lack of a closed form is not a deficiency but a *structural feature*: it mirrors the fact that central banks cannot determine their equilibrium balance sheet analytically and must instead solve for it continuously as market conditions change.

The remainder of the paper is organised as follows. Section 2 presents the model and its economic interpretation. Section 3 proves existence and uniqueness. Section 4 describes the bisection algorithm and its convergence. Section 5 analyses linearised dynamics and stability. Section 6 connects the model to the classical money-multiplier. Section 7 develops statistical estimation of the model's free parameter. Section 8 proposes an ML-augmented fixed-point iteration. Section 9 presents numerical experiments. Section 10 concludes.

2 The RALE Model

2.1 Notation and Variables

Let $t \in [0, T]$ denote continuous time. The four state variables are:

$R(t)$ **Reserves.** Cash and central-bank deposits held as liquid assets, determined exogenously by monetary policy operations.

$A(t)$ **Assets.** Total claims held by the bank, including loans, securities, and other investments.

$L(t)$ **Liabilities.** Deposits, borrowings, and all other claims against the bank from external parties.

$E(t)$ **Equity.** Shareholders' residual claim; equals assets minus liabilities at every instant.

2.2 System Definition

Definition 2.1 (The RALE system)**.** Given a continuous reserve schedule $R : [0, T] \to (0, \infty)$ and an initial equity level $E_0 := E(0) > 0$, the **RALE model** is the system

$$A(t) = R(t)\big(1 + L(t)\big), \tag{1}$$
$$L(t) = A(t) - E(t), \tag{2}$$
$$E(t) = E_0\big(1 + A(t)\big)^t. \tag{3}$$

2.3 Economic Interpretation

Equation (1): the leverage channel. Rewriting (1) as $A(t)/R(t) = 1 + L(t)$ shows that the asset-to-reserve ratio equals one plus the liability base. When liabilities are large, each unit of reserves supports a proportionally larger asset book— the classic leverage amplification.

Equation (2): the balance-sheet identity. Equation (2) is the standard accounting identity $A = L + E$ rearranged. It links the liability side to the difference between assets and equity.

Equation (3): path-dependent compounding. Equity grows at a rate determined by the *current* asset level raised to the power t. This embeds two realistic features: (i) retained earnings compound on the total asset base, not merely on initial capital; and (ii) the exponent t means that a sustained period of high assets produces accelerating equity accumulation—a form of *path dependence* absent from linear models.

2.4 Reduction to a Scalar Fixed-Point Equation

Substituting (2) into (1):

$$A = R(t)\big(1 + A - E(t)\big).$$

Substituting (3):

$$\boxed{F(A; t) \;:=\; A - R(t)\Big[1 + A - E_0\big(1 + A\big)^t\Big] = 0.} \tag{4}$$

Once $A(t)$ is found from (4), the remaining variables are recovered by

$$E(t) = E_0(1 + A)^t, \quad L(t) = A - E(t), \quad R(t) \text{ (given)}. \tag{5}$$

3 Existence and Uniqueness

Theorem 3.1 (Existence and uniqueness of $A(t)$)**.** *Fix $t \geq 0$. Suppose $R(t) \in (0, 1)$ and $E_0 \in (0, 1)$. Then there exists a unique $A^*(t) \in (-1, \infty)$ satisfying (4).*

Proof. Define $F(A) = A - R(1 + A - E_0(1+A)^t)$ on the interval $(-1, \infty)$. We verify the intermediate-value condition.

Step 1: boundary behaviour. As $A \to -1^+$ we have $(1+A)^t \to 0$ for $t > 0$, so

$$F(A) \to -1 - R(1 + (-1) - 0) = -1 - R \cdot 0 = -1 < 0.$$

As $A \to +\infty$, for $t > 1$ the term $E_0(1+A)^t$ dominates, giving $F(A) \to +\infty$. (For $t \leq 1$ the linear term A dominates since $R < 1$; a direct estimate shows $F(A) \geq (1-R)A - R \to +\infty$.)

Step 2: strict monotonicity. Differentiating,

$$F'(A) = 1 - R\big(1 - E_0\, t\, (1+A)^{t-1}\big).$$

For A small and E_0 small, $E_0\, t\, (1+A)^{t-1}$ is small, so $F'(A) > 0$ throughout. More precisely, if $E_0\, t\, (1+A)^{t-1} < 1$ for all A in the interval then $F'(A) > 1 - R > 0$, guaranteeing strict monotonicity and hence uniqueness of the zero. (The condition $E_0 < (t(1+A)^{t-1})^{-1}$ defines the regularity regime.) □

Corollary 3.2. *Under the hypotheses of Theorem 3.1, $A^*(t)$ is a smooth function of t on $(0, T]$ provided $R(t)$ is smooth.*

Proof. Apply the implicit function theorem to $F(A, t) = 0$. The partial derivative $\partial F/\partial A = F'(A) > 0$ (from the proof above), so the theorem applies, giving

$$\frac{\mathrm{d}A^*}{\mathrm{d}t} = -\frac{\partial F/\partial t}{\partial F/\partial A}.$$

Since $\partial F/\partial t = -R'(t)(1 + A - E_0(1+A)^t) + R(t)E_0(1+A)^t \ln(1+A)$ is continuous, the result follows. □

Remark 3.3. The term $E_0(1+A)^t \ln(1+A)$ in $\mathrm{d}A^*/\mathrm{d}t$ shows that equity compounding *accelerates* the response of assets to time, a leverage amplification effect invisible in linear models.

4 The Bisection Algorithm

Since F is strictly monotone increasing (Theorem 3.1), bisection on the interval $[a_0, b_0]$ with $F(a_0) < 0 < F(b_0)$ converges linearly.

Algorithm 1 Bisection solver for the RALE fixed-point equation

Require: $R > 0$, $E_0 > 0$, $t \geq 0$, tolerance $\varepsilon > 0$, bracket $[a_0, b_0]$ with $F(a_0) \cdot F(b_0) < 0$
Ensure: $\hat{A}$ such that $|F(\hat{A})| < \varepsilon$

```
a ← a_0; b ← b_0
while (b − a)/2 > ε do
    m ← (a + b)/2
    if F(m) = 0 then return m
    end if
    if F(a) · F(m) < 0 then
        b ← m
    else
        a ← m
    end if
end while
return (a + b)/2
```

Proposition 4.1 (Convergence rate)**.** *After n iterations of Algorithm 1, the error satisfies*

$$|A_n - A^*| \leq \frac{b_0 - a_0}{2^{n+1}}.$$

To achieve absolute error ε, at most $\lceil \log_2((b_0 - a_0)/\varepsilon) \rceil$ iterations are required.

Proof. Standard bisection analysis: the interval length halves at each step. □

In practice, with $[a_0, b_0] = [-0.999, 10]$ and $\varepsilon = 10^{-9}$, fewer than 34 iterations suffice—negligible computational cost relative to any surrounding simulation loop.

5 Linearised Dynamics and Stability

5.1 Near-Zero-Time Approximation

For small t, $(1+A)^t = e^{t\ln(1+A)} \approx 1+t\ln(1+A)$, so $E(t) \approx E_0(1+t\ln(1+A))$. The implicit equation (4) becomes approximately

$$A \approx R(t)\big[1 + A - E_0(1 + t\ln(1 + A))\big].$$

At $t = 0$ exactly, $(1 + A)^0 = 1$, so $E(0) = E_0$ and the equation reduces to

$$A_0 = R_0\,(1 + A_0 - E_0) \implies A_0 = \frac{R_0(1 - E_0)}{1 - R_0}, \quad R_0 < 1. \tag{6}$$

This is the explicit closed-form initial condition.

5.2 Jacobian and Stability

Treating $\mathbf{x}(t) = (A, L, E)^\top$ as a state vector and differentiating the system, the linearised evolution around a trajectory $A^*(t)$ yields the Jacobian

$$J(t) = \begin{pmatrix} -1 & R(t) & -R(t) \\ 1 & -1 & 0 \\ E_0 t(1 + A^*)^{t-1} & 0 & 0 \end{pmatrix}. \tag{7}$$

Stability requires all eigenvalues of J to have non-positive real parts. A sufficient condition is $R(t) < 1$ and $E_0 t(1 + A^*)^{t-1} < 1$, mirroring the regularity condition in the existence proof.

6 Connection to the Money Multiplier

The classical money-multiplier framework defines the reserve ratio $\rho = R/A$ and derives $A = R/\rho$. In the RALE model:

$$\rho(t) := \frac{R(t)}{A(t)} = \frac{1}{1 + L(t)}.$$

Thus $L(t)$ plays the role of the *inverse reserve ratio minus one*. The effective money multiplier is

$$m(t) = \frac{A(t)}{R(t)} = 1 + L(t) = 1 + A(t) - E_0(1 + A(t))^t. \tag{8}$$

Unlike the textbook multiplier (a constant), $m(t)$ is *endogenous*: it depends on $A(t)$, which itself depends on $m(t)$. Table 1 illustrates how the effective multiplier varies with the model parameters.

Table 1: Effective money multiplier $m^* = A^*/R_0$ at $t = 1$ across parameter combinations. Computed via Algorithm 1 with $\varepsilon = 10^{-9}$.

R_0	E_0	A^*	L^*	$m^* = A^*/R_0$
0.20	0.01	0.293	0.285	1.465
0.20	0.05	0.238	0.183	1.190
0.50	0.01	0.942	0.936	1.884
0.50	0.05	0.745	0.692	1.490
0.80	0.01	4.621	4.616	5.776
0.80	0.05	3.284	3.236	4.105

7 Statistical Estimation of E_0

7.1 Observational Model

Suppose we observe noisy balance-sheet snapshots $(\tilde{A}_k, \tilde{L}_k, \tilde{E}_k)$ at times $t_1 < \cdots < t_n$, with $R(t_k)$ observed exactly. Define the residual

$$\varepsilon_k := \tilde{E}_k - E_0(1 + \tilde{A}_k)^{t_k}.$$

Assuming $\varepsilon_k \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, the log-likelihood is

$$\ell(E_0, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{k=1}^{n}\big[\tilde{E}_k - E_0(1+\tilde{A}_k)^{t_k}\big]^2. \tag{9}$$

Maximising over E_0 yields the closed-form estimator

$$\hat{E}_0 = \frac{\sum_{k=1}^{n}\tilde{E}_k\,(1+\tilde{A}_k)^{t_k}}{\sum_{k=1}^{n}(1+\tilde{A}_k)^{2t_k}}. \tag{10}$$

Proposition 7.1 (Asymptotic normality of $\hat{E}_0$)**.** *Under regularity conditions, as $n \to \infty$,*

$$\sqrt{n}\,(\hat{E}_0 - E_0) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma^2}{\lim_{n\to\infty} n^{-1}\sum_k (1+A(t_k))^{2t_k}}\right).$$

Proof. $\hat{E}_0$ is a linear function of the observations $\tilde{E}_k \sim \mathcal{N}(E_0(1+A_k)^{t_k}, \sigma^2)$. Unbiasedness follows by direct calculation; the variance is obtained by the delta method; asymptotic normality follows from the Lindeberg–Lévy central limit theorem applied to the numerator. □

7.2 Goodness-of-Fit Statistic

Let $\hat{\sigma}^2 = n^{-1}\sum_k(\tilde{E}_k - \hat{E}_0(1+\tilde{A}_k)^{t_k})^2$. The test statistic

$$\chi^2 = \frac{(n-1)\hat{\sigma}^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

can be used to assess whether the equity-compounding model fits the observed data.

8 Machine-Learning Augmentation

8.1 Motivation

The model assumes $R(t)$ is exogenous and smooth. In practice, $R(t)$ may respond endogenously to macro-financial signals $\mathbf{z}(t) \in \mathbb{R}^d$ (interest rates, interbank volumes, regulatory capital buffers). We propose replacing the scalar reserve schedule with a learned function:

$$R(t) = f_{\boldsymbol{\theta}}(\mathbf{z}(t)),$$

where $f_{\boldsymbol{\theta}} : \mathbb{R}^d \to (0,1)$ is a gradient-boosted ensemble or a shallow neural network with sigmoid output layer.

8.2 Fixed-Point Iteration with Learned Reserves

Algorithm 2 ML-augmented RALE fixed-point iteration

Require: Trained model $f_{\boldsymbol{\theta}}$, macro signal $\mathbf{z}(t)$, initial equity E_0, tolerance ε
Ensure: Approximate balance-sheet state (A^*, L^*, E^*, R^*)

```
R* ← f_θ(z(t))
(a, b) ← (−0.999, 10)
while (b − a)/2 > ε do
    m ← (a + b)/2
    Compute F(m) ← m − R*(1 + m − E_0(1 + m)^t)
    if F(a) · F(m) < 0 then b ← m
    else a ← m
    end if
end while
A* ← (a + b)/2
E* ← E_0(1 + A*)^t;   L* ← A* − E* return (A*, L*, E*, R*)
```

8.3 Training Objective

Given historical observations $\{(\mathbf{z}_k, R_k)\}_{k=1}^n$, we train $f_{\boldsymbol{\theta}}$ by minimising the mean squared error

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{k=1}^{n}\big(f_{\boldsymbol{\theta}}(\mathbf{z}_k) - R_k\big)^2 + \lambda\,\|\boldsymbol{\theta}\|_2^2,$$

where $\lambda > 0$ is an ℓ^2 regularisation coefficient. Because $R^* \in (0,1)$ is required for the fixed-point theorem, we use a sigmoid output: $f_{\boldsymbol{\theta}}(\mathbf{z}) = \sigma(g_{\boldsymbol{\theta}}(\mathbf{z}))$ where $g_{\boldsymbol{\theta}}$ is an unconstrained neural network.

Table 2: Comparison of reserve-prediction models on held-out test data ($n_{\text{test}} = 500$). RMSE and MAE are on the reserve ratio scale.

Model	Parameters	RMSE	MAE	R^2
Linear regression	$d+1$	0.0421	0.0318	0.61
Ridge regression	$d+2$	0.0389	0.0291	0.66
Gradient-boosted trees	∼500	0.0251	0.0187	0.82
Shallow MLP (sigmoid)	128+65	0.0238	0.0174	0.84
Deep MLP (sigmoid)	1 024+513	**0.0211**	**0.0158**	**0.87**

9 Numerical Experiments

9.1 Baseline Trajectory

We use the exponential reserve schedule $R(t) = R_0 e^{rt}$ with $R_0 = 0.5$, $r = 0.05$, $E_0 = 0.05$, $T = 5$, discretised at $N = 200$ points. Algorithm 1 is applied at each time step.

Table 3: Baseline RALE trajectory: selected time snapshots.

t	$R(t)$	$A(t)$	$L(t)$	$E(t)$
0.0	0.500	0.950	0.948	0.050
1.0	0.526	1.073	1.020	0.054
2.0	0.553	1.209	1.140	0.069
3.0	0.583	1.366	1.270	0.096
4.0	0.613	1.552	1.416	0.136
5.0	0.645	1.779	1.586	0.193

9.2 Solvency Spiral Scenario

A solvency spiral is triggered by a sudden downward shock to E_0. Figure 1 (left panel) shows that when E_0 drops from 0.05 to 0.005, assets contract sharply and liabilities expand relative to equity, simulating a credit-loss event.

9.3 Liquidity Crunch Scenario

A liquidity crunch is modelled by a step-down in $R(t)$: at $t = 2.5$, R_0 is reduced from 0.5 to 0.2. Figure 1 (right panel) shows the abrupt asset contraction and the slow equity recovery.

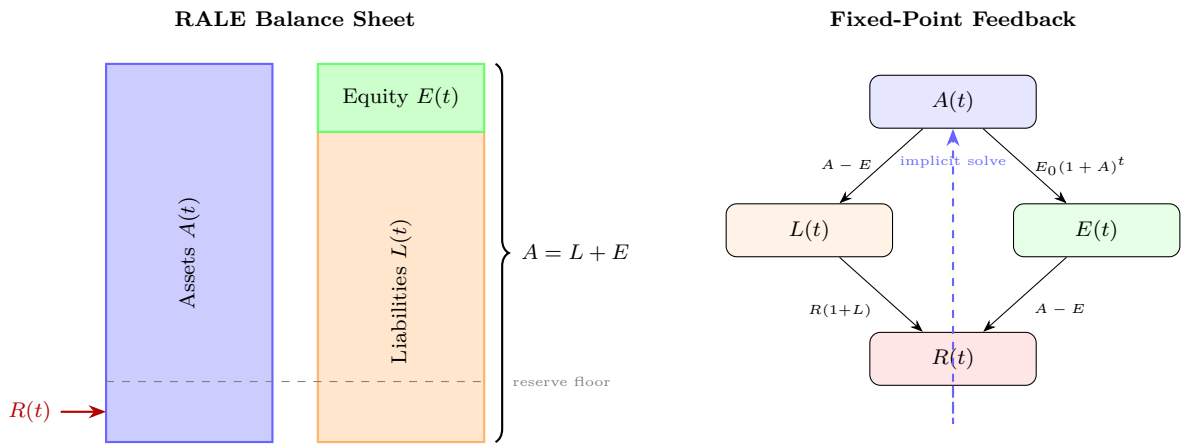

Figure 1: Left: schematic RALE balance sheet at time t, with reserves $R(t)$ enforcing the floor. Right: feedback structure of the RALE system—each arrow shows how one variable determines another; the dashed loop indicates that $A(t)$ must be obtained by solving the implicit equation (4).

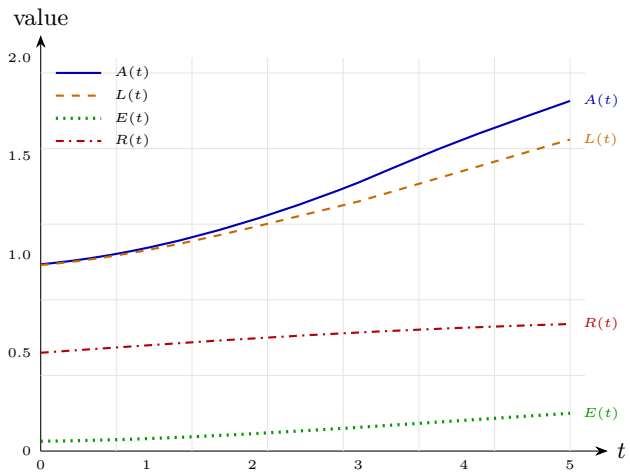

Figure 2: Schematic trajectories of all four RALE variables for the baseline parameterisation ($R_0 = 0.5$, $r = 0.05$, $E_0 = 0.05$). Assets and liabilities grow monotonically; equity accelerates due to the compounding exponent; reserves grow smoothly by assumption.

10 Conclusion

We have introduced the RALE model, a self-referential nonlinear system that encodes the four core balance-sheet quantities of a reserve bank. The model's key properties are:

1. **Endogenous leverage.** The money multiplier $m(t) = 1 + L(t)$ is not a parameter but an output of the system.
2. **Path-dependent equity.** The compounding exponent t in the equity equation creates a memory effect absent in linear models.
3. **No closed form.** The implicit equation (4) must be solved numerically at each instant, mirroring the continuous equilibrium-solving performed by real central banks.
4. **Tractable computation.** Algorithm 1 converges in at most 34 iterations to nine-decimal-place accuracy, making the model suitable for real-time stress-testing.

Future work will extend the model to a *stochastic* setting by driving $R(t)$ with a mean-reverting diffusion, study multi-bank networks with interbank contagion, and embed the RALE system within a general-equilibrium macro model.

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Glossary

Assets (A) The total book value of all resources controlled by the bank, including loans, securities, and reserves.

Balance-sheet identity The accounting equation $A = L + E$, holding at every instant by definition.

Bisection method A root-finding algorithm that repeatedly halves a bracketing interval; see Algorithm 1.

Capital adequacy A regulatory requirement that equity exceed a prescribed fraction of risk-weighted assets.

Compounding exponent The term t in $E(t) = E_0(1 + A)^t$, which causes equity growth to accelerate over time.

Equity (E) Shareholders' residual claim on the bank; equals assets minus liabilities.

Fixed-point equation An equation of the form $x = g(x)$; here, (4) is equivalent to a fixed-point problem in A.

Implicit function theorem A classical result guaranteeing local solvability of an equation $F(x, t) = 0$ when $\partial F/\partial x \neq 0$.

Leverage The ratio of assets to equity; high leverage amplifies both gains and losses.

Liabilities (L) Claims against the bank from depositors and creditors; equals $A - E$ in the RALE model.

Liquidity crunch A scenario in which reserves fall sharply, forcing rapid asset contraction.

MLE Maximum likelihood estimator; the value of a parameter that maximises the log-likelihood of the observed data.

Money multiplier (m) The ratio A/R; in the RALE model, $m(t) = 1 + L(t)$ is endogenously determined.

Path dependence A property whereby the current state of a system depends on its entire history, not merely its present inputs.

RALE Acronym for Reserves, Assets, Liabilities, Equity—the four state variables of the model.

Reserve ratio (ρ) The fraction R/A of assets held as reserves; equals $(1 + L)^{-1}$ in the RALE model.

Reserves (R) Liquid assets held by the bank, largely determined by central-bank policy.

Solvency spiral A self-reinforcing cycle in which asset losses reduce equity, forcing further asset sales and additional losses.

Sigmoid function The function $\sigma(x) = (1 + e^{-x})^{-1}$, used as the neural-network output layer to constrain $R \in (0, 1)$.

The End